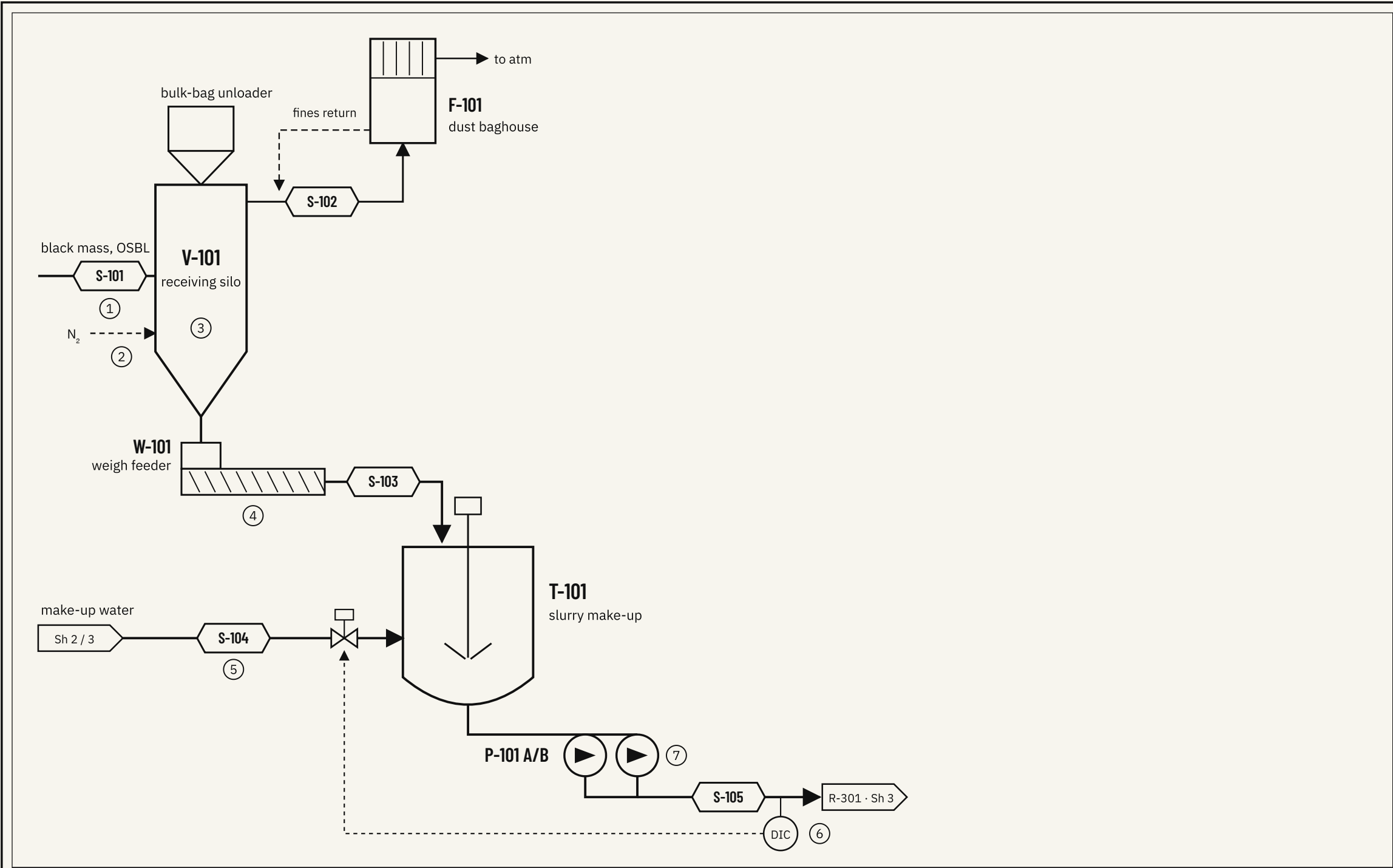
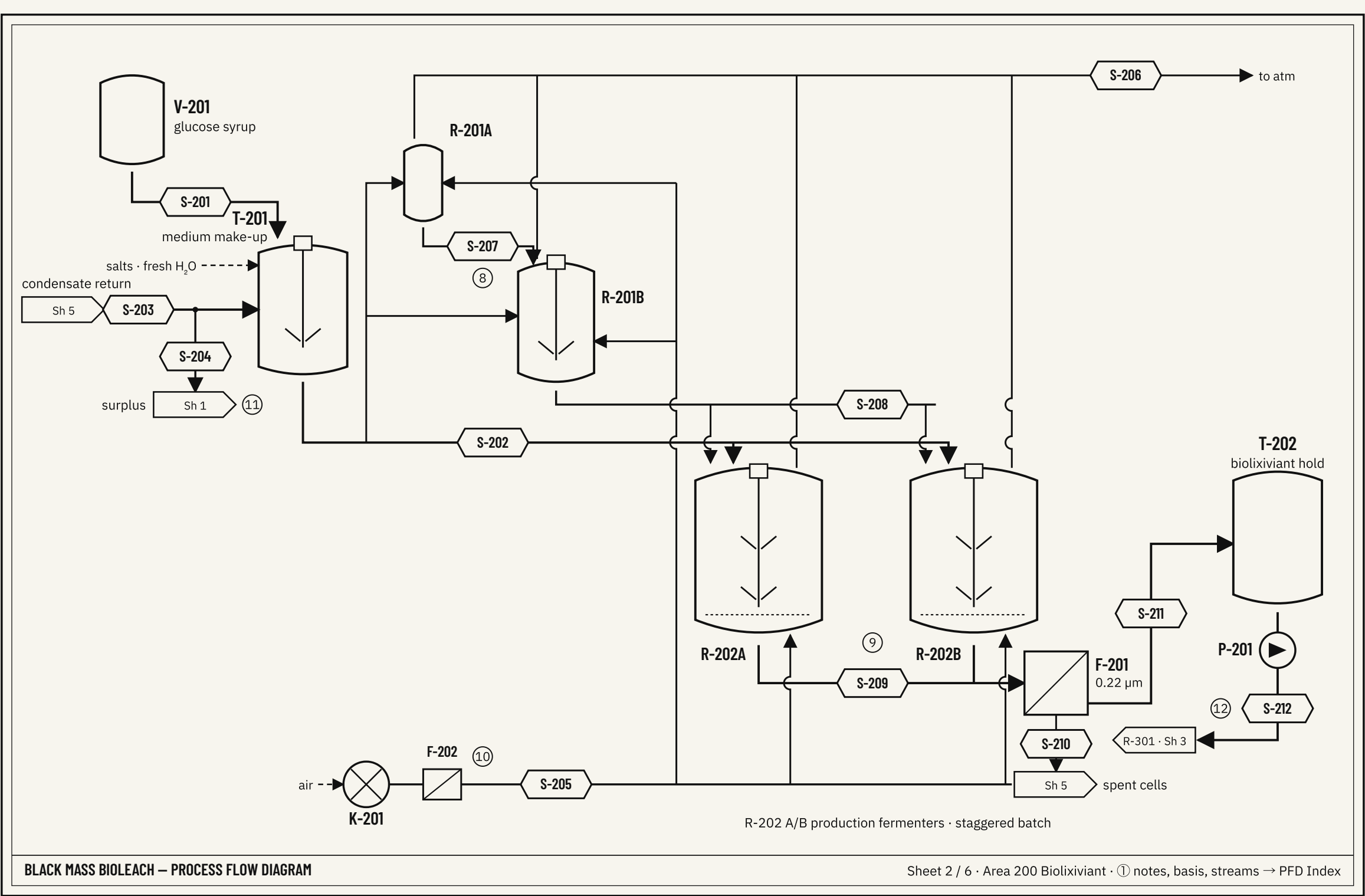


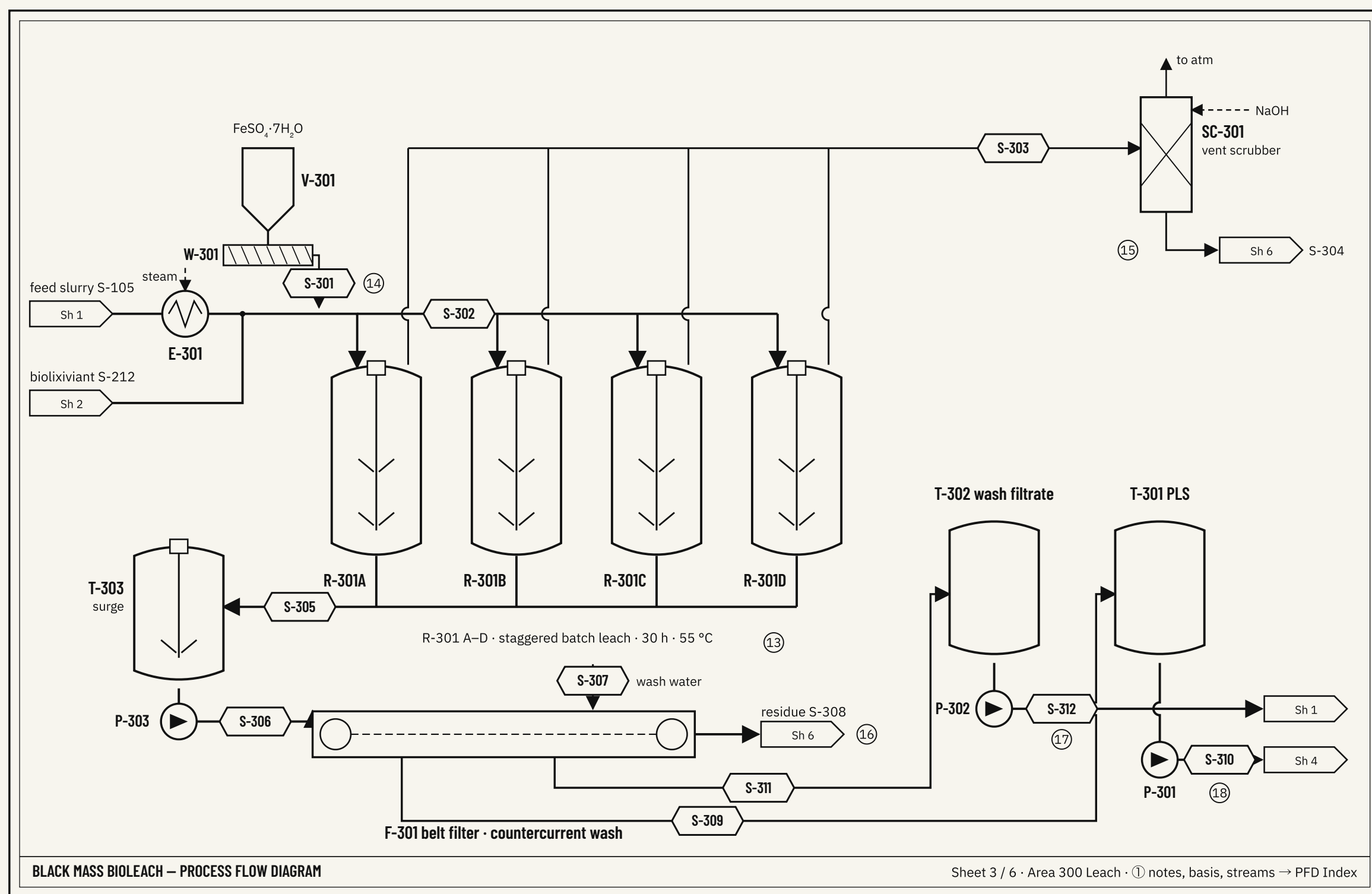
Black Mass Bioleach PFD

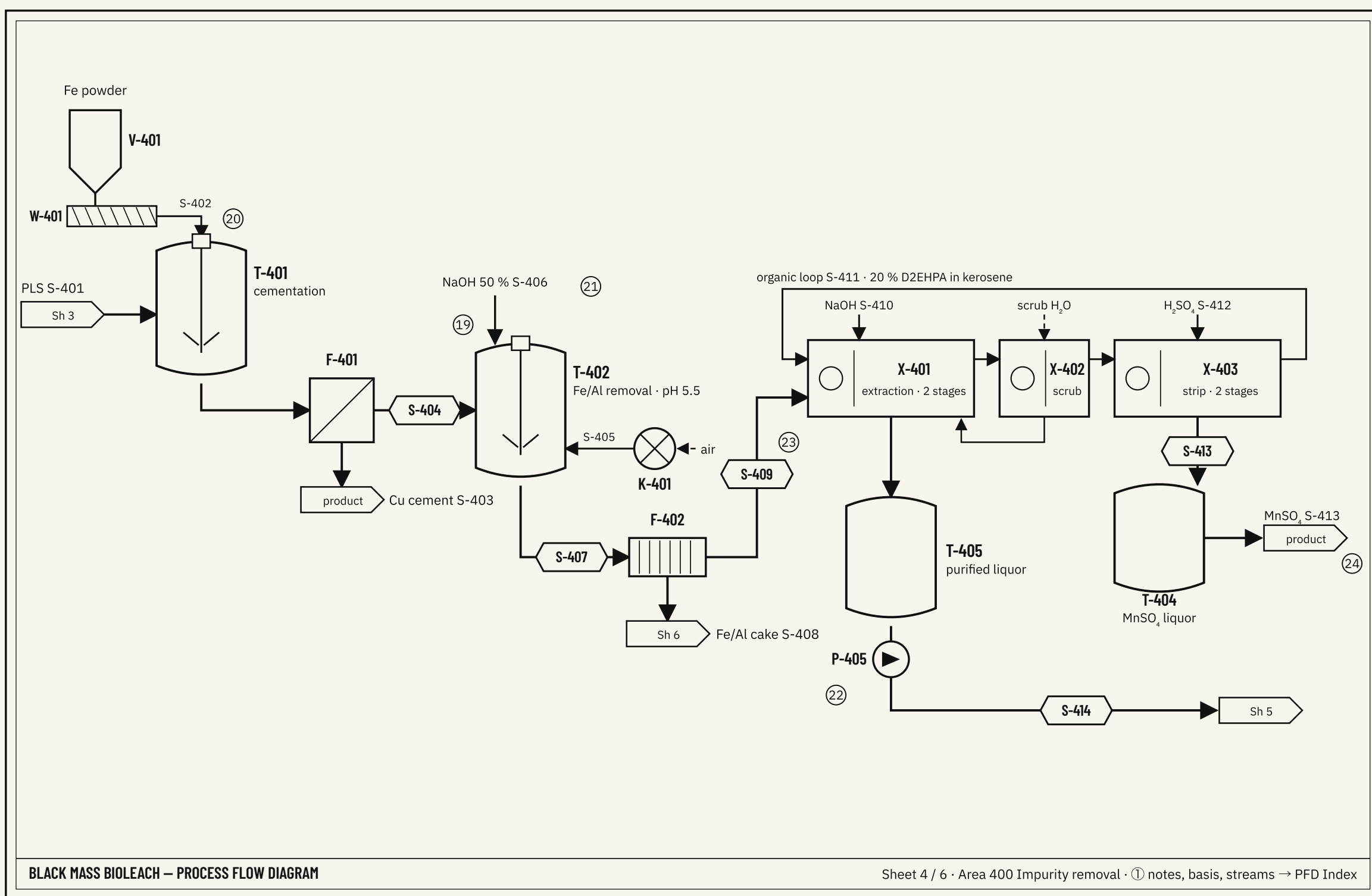
Live drawing, added to one block at a time. Equipment and stream detail is kept in the companion PFD Index.

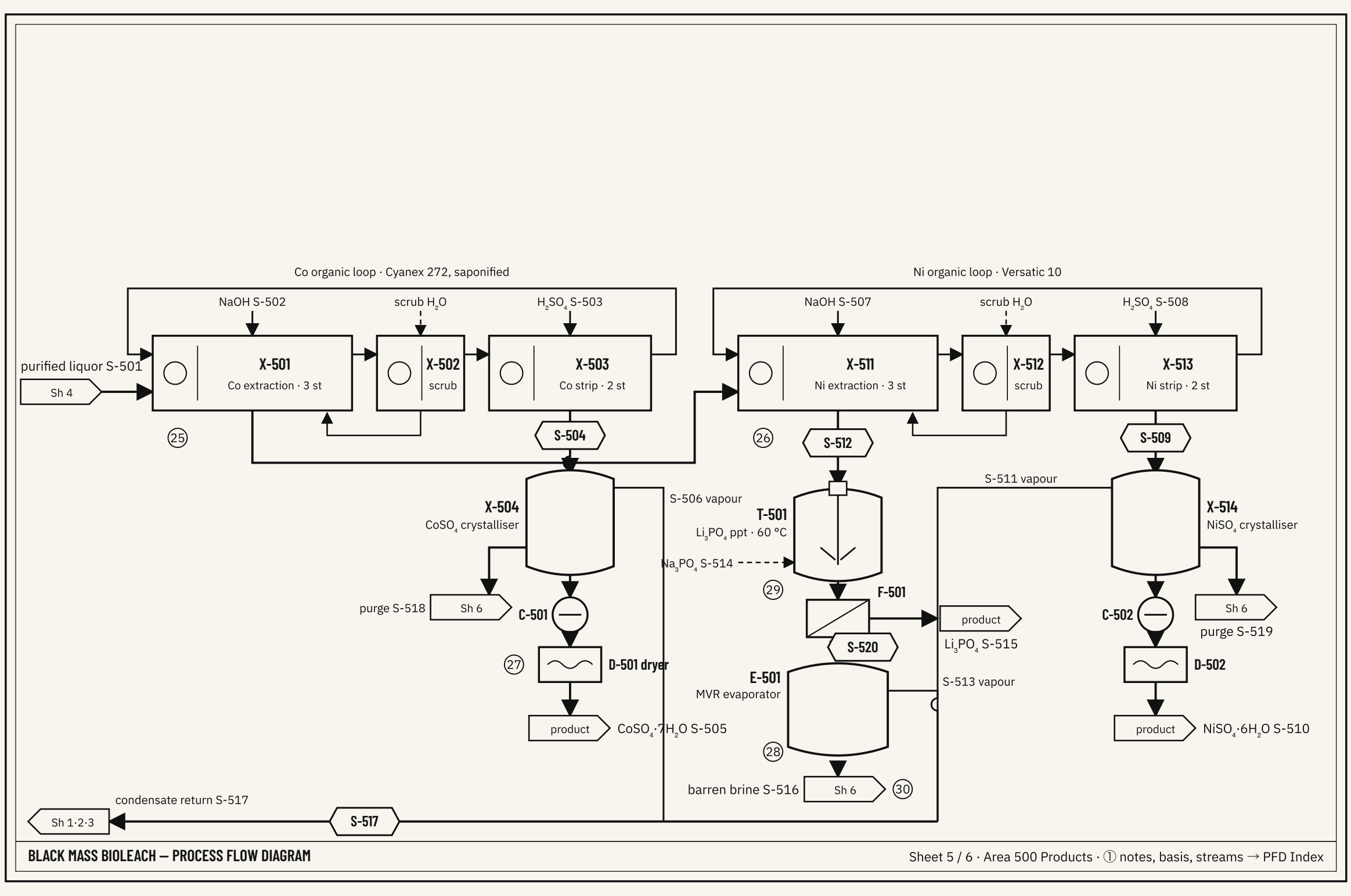
Sheet 1 · Area 100 · Feed preparation



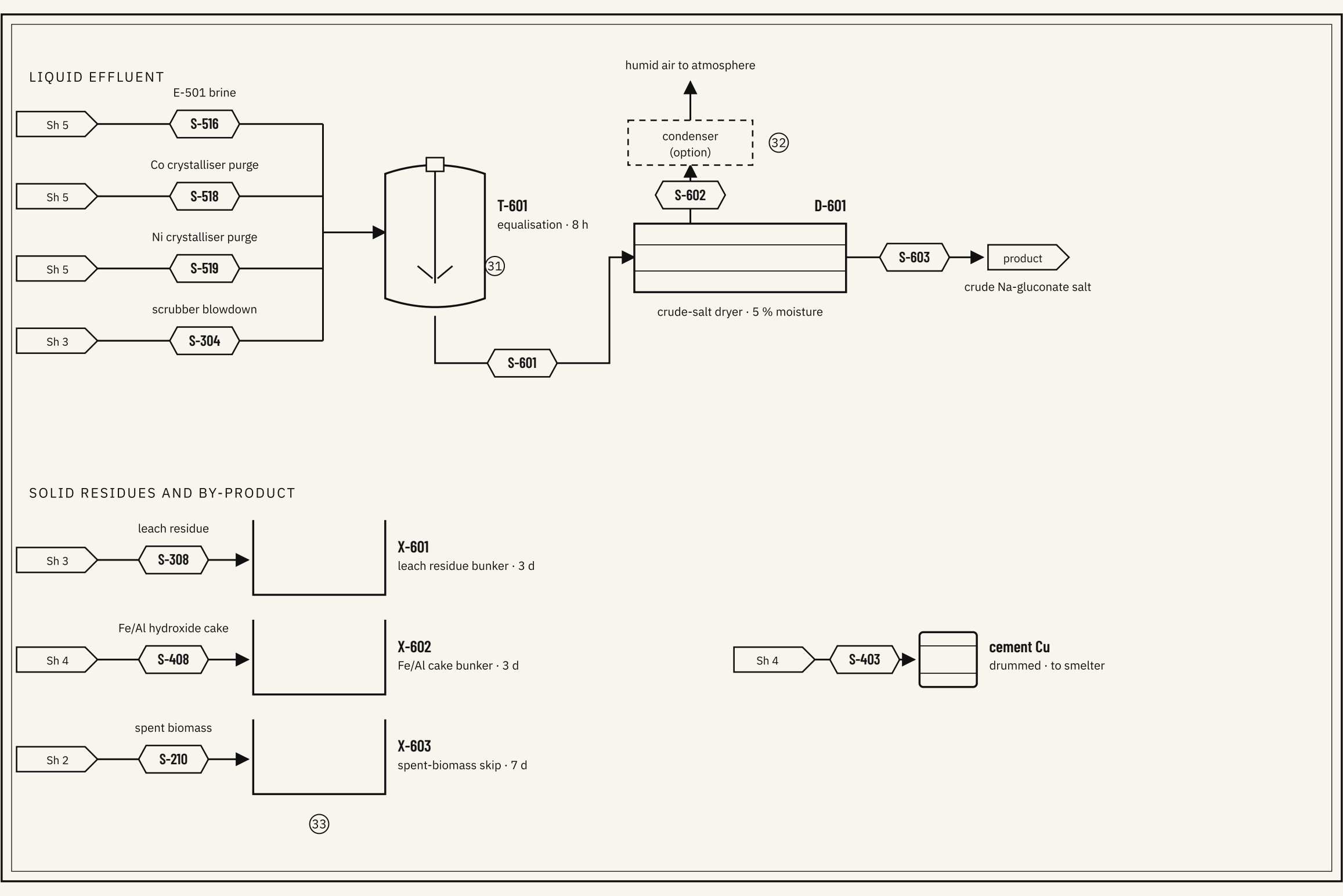








BLACK MASS BIOLEACH – PROCESS FLOW DIAGRAM



Stream routing

NO.	FROM	TO	CARRIES
S-101	Battery pretreatment (OSBL)	V-101	black mass
S-102	V-101 vent	F-101	N ₂ / air + fines
S-103	W-101	T-101	metered black mass
S-104	Sheet 2 (condensate surplus S-204) + Sheet 3 (wash filtrate S-312) + fresh	T-101	make-up water
S-105	P-101 A/B	R-301 (Sheet 3)	feed slurry
S-201	V-201	T-201	glucose syrup
S-202	T-201	R-201 A/B, R-202 A/B	fermentation medium
S-203	Sheet 5 (condensate S-517)	T-201	returned water
S-204	S-203 tee	Sheet 1 (T-101)	condensate surplus
S-205	K-201 / F-202	all fermenters	sterile air
S-206	all fermenters	atmosphere	off-gas
S-207	R-201A	R-201B	inoculum (per batch)
S-208	R-201B	R-202 A or B	inoculum (per batch)
S-209	R-202 A/B	F-201	harvested broth
S-210	F-201	Sheet 6 (X-603)	spent cells
S-211	F-201	T-202	cell-free biolixiviant
S-212	P-201	R-301 (Sheet 3)	biolixiviant dose
S-301	V-301 / W-301	R-301 fill line	FeSO ₄ ·7H ₂ O, dry
S-302	E-301 + S-212 tee	R-301 A–D	heated feed slurry + biolixiviant
S-303	R-301 A–D	SC-301	off-gas (CO ₂ , HF traces)
S-304	SC-301	Sheet 6 (T-601)	scrubber blowdown
S-305	R-301 A–D	T-303	leached slurry (batch drain)
S-306	P-303	F-301	leached slurry (continuous)
S-307	process water	F-301	cake wash
S-308	F-301	Sheet 6	leach residue (graphite, binder, unleached cathode)
S-309	F-301 strong filtrate	T-301	pregnant leach solution
S-310	P-301	Sheet 4	PLS to Fe removal
S-311	F-301 wash filtrate	T-302	weak liquor
S-312	P-302	Sheet 1 (T-101, part of S-104)	wash filtrate recycle

NO.	FROM	TO	CARRIES
S-401	Sheet 3 (S-310)	T-401	PLS
S-402	V-401 / W-401	T-401	iron powder
S-403	F-401	product	cement copper
S-404	F-401	T-402	Cu-free liquor
S-405	K-401	T-402	oxidation air
S-406	NaOH 50 %	T-402	neutralisation to pH 5.5
S-407	T-402	F-402	goethite / gibbsite slurry
S-408	F-402	Sheet 6	Fe/Al hydroxide cake
S-409	F-402	X-401	Fe/Al-free liquor
S-410	NaOH 50 %	X-401	SX pH control
S-411	X-401 → X-402 → X-403 → X-401	internal	organic loop (D2EHPA)
S-412	H ₂ SO ₄ 98 %	X-403	strip acid
S-413	X-403 → T-404	product	MnSO ₄ strip liquor
S-414	P-405	Sheet 5	purified Co/Ni/Li liquor
S-501	Sheet 4 (S-414)	X-501	purified liquor
S-502 / S-507	NaOH 50 %	X-501 / X-511	SX pH control
S-503 / S-508	H ₂ SO ₄ 98 %	X-503 / X-513	strip acid
S-504	X-503	X-504	CoSO ₄ strip liquor
S-505	D-501	product	CoSO ₄ ·7H ₂ O
S-506 / S-511 / S-513	X-504 / X-514 / E-501	condensate header	vapour → condensate
S-509	X-513	X-514	NiSO ₄ strip liquor
S-510	D-502	product	NiSO ₄ ·6H ₂ O
S-512	X-511 raffinate	T-501	Li / Na / gluconate liquor
S-514	Na ₃ PO ₄	T-501	phosphate reagent
S-515	F-501	product	Li ₃ PO ₄
S-520	F-501 filtrate	E-501	Li-free Na-gluconate liquor
S-516	E-501	Sheet 6	barren Na-gluconate brine (the bleed)
S-517	condensate header	Sheets 1 · 2 · 3	clean water return
S-518 / S-519	X-504 / X-514	Sheet 6	crystalliser mother-liquor purge
S-601	T-601	D-601	equalised effluent (Na-gluconate / phosphate / sulfate)

NO.	FROM	TO	CARRIES
S-602	D-601	atmosphere	humid dryer exhaust — condensable (note ③2)
S-603	D-601	OSBL by-product	crude sodium gluconate salt, 5 % moisture

Flows, compositions, and equipment sizing live in the [PFD Index](#), one section per block.
This page is only meant to show where material goes.